2025-2026 Precalculus Scope and Sequence

High School Semester Block

Refer to Unit Planning Organizers for Instructional Guidance

Unit	Unit Title	S1: ≈ # of Weeks	S1: Date Range	S2: ≈ # of Weeks	S2: Date Range
1	***Incorporate WIM - Week of Inspirational Math*** **Unit 1: Building Mathematical Community w/Parent Functions & Key Features** ***(Review NC Math 3 Standards)***	1 week	8/13 – 8/19	1 week	1/5 – 1/9
2	Extending Polynomial and Rational Functions	4 weeks	8/20 – 9/17	4 weeks	1/12 – 2/9
3	Composition and Inverse Functions	3 weeks	9/18 – 10/8	3 weeks	2/10 –3/3
4	Exponential & Logarithmic Functions	4 weeks	10/9 – 11/10	4 weeks	3/4– 3/31
5	Trigonometry	4 weeks	11/12 – 12/2	4 weeks	4/1 – 4/27
6	Parametric Equations	1 week	12/3 – 12/5	1 week	4/28 – 5/1
7	ACT Prep - Incorporated Earlier	N/A	N/A	N/A	N/A
	Review **Teacher Made Exam**		**12/8– 12/12** **12/15 – 12/19**		**5/4 – 5/8** **5/11– 5/22**

The number of suggested weeks for each unit includes assessment and review days.
Each unit will need flexibility based on the need for reteaching and remediation.

Standards for Mathematical Practice	
1. Make sense of problems and persevere in solving them.	5. Use appropriate tools strategically.
2. Reason abstractly and quantitatively.	6. Attend to precision.
3. Construct viable arguments and critique reasoning of others.	7. Look for and make use of structure.
4. Model with mathematics	8. Look for and express regularity in repeated reasoning.

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Unit 1: Building Mathematical Community w/Parent Functions & Key Features		**≈ # of Weeks:** **S1: 1 week / S2: 1 week**
Interpreting Functions		
NC.M3.F-IF.4	Interpret key features of graphs, tables, and verbal descriptions in context to describe functions that arise in applications relating two quantities to include periodicity and discontinuities.	
Building Functions		
NC.M3.F-BF.3	Extend an understanding of the effects on the graphical and tabular representations of a function when replacing $f(x)$ with $k{\cdot}f(x)$, $f(x)+k$, $f(x+k)$ to include $f(k{\cdot}x)$ for specific values of k (both positive and negative).	

Unit 2: Extending Polynomial and Rational Functions		**≈ # of Weeks:** **S1: 4 weeks / S2: 4 weeks**
Solving Rational and Polynomial Inequalities		
NC.PC.A.1.2	Implement graphical methods to solve rational and polynomial inequalities.	
NC.PC.A.1.1	Implement algebraic (sign analysis) methods to solve rational and polynomial inequalities.	
Algebraic and Graphical Representations of Rational and Polynomial Functions		
NC.PC.F.4.6	Implement graphical and algebraic methods to solve optimization problems given rational and polynomial functions in context with support from technology.	

NC.PC.F.4.5	Interpret algebraic and graphical representations to determine key features of rational functions. Key features include: domain, range, intercepts, intervals where the function is increasing, decreasing, positive or negative, concavity, end behavior, continuity, limits, and asymptotes.
NC.PC.F.4.8	Identify the conic section (ellipse, hyperbola) from its algebraic representation in standard form
NC.PC.F.4.9	Interpret algebraic and graphical representations to determine key features of conic sections (ellipse: center, length of the major and minor axes; hyperbola: vertices, transverse axis; parabola: vertex, axis of symmetry).
Properties of Complex Numbers and the Complex Number System *(ACT Standards)*	
NC.PC.N.1.1	Execute the sum and difference algorithms to combine complex numbers.
NC.PC.N.1.2	Execute the multiplication algorithm with complex numbers.

Unit 3: Composition and Inverse Functions	**≈ # of Weeks:** **S1: 3 weeks / S2: 3 weeks**
Properties of Function Composition & Building New Functions	
NC.PC.F.5.1	Implement algebraic procedures to compose functions.
NC.PC.F.5.2	Execute a procedure to determine the value of a composite function at a given value using algebraic, graphical, and tabular representations.
NC.PC.F.5.3	Implement algebraic methods to find the domain of a composite function.
NC.PC.F.5.4	Organize information to build models involving function composition.

NC.PC.F.5.5	Deconstruct a composite function into two functions.
NC.PC.F.5.6	Implement algebraic and graphical methods to find an inverse function of an existing function, restricting domains if necessary
NC.PC.F.5.7	Use composition to determine if one function is the inverse of another function.
Matrices *(ACT Standards)*	
NC.PC.N.2.1	Execute the sum and difference algorithms to combine matrices of appropriate dimensions.
NC.PC.N.2.2	Execute associative and distributive properties to matrices.
NC.PC.N.2.3	Execute commutative property to add matrices.
NC.PC.N.2.4	Execute properties of matrices to multiply a matrix by a scalar.
NC.PC.N.2.5	Execute the multiplication algorithm with matrices

Unit 4: Exponential & Logarithmic Functions	**≈ # of Weeks: S1: 4 weeks / S2: 4 weeks**
Recursive Functions	
NC.PC.F.6.1	Use algebraic representations to build recursive functions.
NC.PC.F.6.2	Construct a recursive function for a sequence represented numerically.
Algebraic and Graphical Representations of Exponential & Logarithmic Functions	

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NC.PC.F.4.2	Integrate information to build exponential functions to model phenomena involving growth or decay.
NC.PC.F.4.1	Interpret algebraic and graphical representations to determine key features of exponential functions. *Key features include: domain, range, intercepts, intervals where the function is increasing, decreasing, positive or negative, concavity, end behavior, limits, and asymptotes.*
NC.PC.F.4.7	Construct graphs of transformations of power, **exponential**, and **logarithmic** functions showing key features.
NC.PC.F.4.3	Interpret algebraic and graphical representations to determine key features of logarithmic functions. *Key features include: domain, range, intercepts, intervals where the function is increasing, decreasing, positive or negative, concavity, end behavior, continuity, limits, and asymptotes.*
NC.PC.F.4.4	Implement graphical and algebraic methods to solve exponential and logarithmic equations in context with support from technology.
Properties of Exponentials & Logarithms	
NC.PC.A.2.1	Use properties of logarithms to rewrite expressions.
NC.PC.A.2.2	Implement properties of exponentials and logarithms to solve equations.
Composition & Inverse Functions	
NC.PC.F.5.1	Implement algebraic procedures to compose functions.
NC.PC.F.5.6	Implement algebraic and graphical methods to find an inverse function of an existing function, restricting domains if necessary
NC.PC.F.5.7	Use composition to determine if one function is the inverse of another function.

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Unit 5: Trigonometry		**≈ # of Weeks:** **S1: 4 weeks / S2: 4 weeks**
Properties of a Unit Circle & Trig Values		
NC.PC.F.2.2	Explain how the symmetry of the unit circle is related to the periodicity of trigonometric functions	
NC.PC.F.2.1	Use a unit circle to find values of sine, cosine, and tangent for angles in terms of reference angles.	
NC.PC.F.2.3	Implement properties of trigonometric functions to solve equations including inverse trigonometric functions, double angle formulas, and Pythagorean identities.	
Key Features of Trig Functions		
NC.PC.F.1.1	Interpret algebraic and graphical representations to determine key features of transformed sine and cosine functions. Key features include: amplitude, domain, midline, phase shift, frequency, period, intervals where the function is increasing, decreasing, positive or negative, relative maximums and minimums.	
NC.PC.F.1.3	Integrate information to build trigonometric functions with specified amplitude, frequency, period, phase shift, or midline with or without context.	
NC.PC.F.1.4	Implement graphical and algebraic methods to solve trigonometric equations and inequalities in context with support from technology.	
NC.PC.F.1.2	Interpret algebraic and graphical representations to determine key features of tangent, cotangent, secant, and cosecant. Key features include: domain, frequency, period, intervals where the function is increasing, decreasing, positive or negative, relative maximums and minimums, and asymptotes.	
Properties of a Unit Circle & Trig Values		
NC.PC.F.3.3	Implement the Pythagorean identity to find sin(θ), cos(θ), or tan(θ) given sin(θ), cos(θ), or tan(θ) and the quadrant of the	

	angle.
NC.PC.F.3.1	Implement a strategy to solve equations using inverse trigonometric functions.
NC.PC.F.3.2	Implement Law of Sines and Law of Cosines to solve problems.
Finding Inverse Functions	
NC.PC.F.5.6	Implement algebraic and graphical methods to find an inverse function of an existing function, restricting domains if necessary.
Understanding Properties & Operations with Vectors (ACT Standards)	
NC.PC.N.3.1	Represent a vector indicating magnitude and direction.
NC.PC.N.3.2	Execute sum and difference algorithms to combine vectors

Unit 6: Parametric Equations	**≈ # of Weeks:** **S1: 1 week / S2: 1 week**
Review / Teacher Made Exam	
Building Parametric Functions	
NC.PC.F.7.1	Implement algebraic methods to write parametric equations in context.
NC.PC.F.7.2	Implement technology to solve contextual problems involving parametric equations.
Rewriting Parametric Equations	
NC.PC.A.2.4	Implement algebraic techniques to rewrite parametric equations in cartesian form by eliminating the parameter.

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